



# **DAN KAMAU MWAURA**

## **Software Engineer | Full-Stack Mobile & AI Agent Systems |**

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## **PROFESSIONAL SUMMARY**

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Self-taught software engineer with a proven track record shipping full-stack Flutter applications backed by production-grade databases, now redirecting that engineering discipline toward AI agent automation. Combines hands-on experience in mobile/desktop app architecture (Dart, Flutter, Serverpod, Supabase) with an intensive, self-directed research program in multi-agent orchestration (CrewAI, LangGraph). Reads source documentation cover-to-cover and dissects open-source codebases rather than following tutorials — an engineer who builds first and debugs from the stack trace.

## **TECHNICAL SKILLS**

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- **Languages & Frameworks:** | Dart, Flutter, Serverpod | Python | Typescript | Javascript
- **Databases & Backend-as-a-Service:** PostgreSQL, Firebase, Supabase, MariaDB
- **AI & Automation (Current Focus):** AI Agents, CrewAI, LangGraph, Agentic Workflow Design
- **Engineering:** Relational data modelling, API intergration, authentication, database migrations, async programming, tool calling
- **Cross-platform building:** saving time using one code base across different platforms and systems(linux, windows,mobile, web, mac)
- **State Management:** Provider, Riverpod, Bloc/Cubic, GetX, Global State, Server State

## FEATURED ENGINEERING PROJECTS

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### Full-Stack Farm/Field Management App: Flutter · Serverpod · PostgreSQL · Supabase

- Architected a cross-platform mobile/desktop client with a Serverpod backend, replacing REST-only calls with typed, code-generated client-server calls to eliminate a class of serialization bugs.
- Modeled a normalized PostgreSQL schema for time-series operational data, enabling sub-second historical queries across thousands of records without denormalized caching.
- Implemented reactive state management to keep offline-first mobile clients synchronized with the server on reconnect, resolving conflicts deterministically.

### AI Agent Automation Pipeline Python | CrewAI | LangGraph

- Designed a multi-agent workflow using CrewAI to decompose a single automation task into specialized roles (research, synthesis, execution), each with scoped tool access to reduce hallucinated actions.
- Built a LangGraph state machine to control agent handoffs explicitly, replacing implicit prompt-chaining with auditable, resumable execution graphs.
- Instrumented the pipeline with structured logging at each node transition, cutting debugging time on multi-step agent failures by making intermediate reasoning inspectable.

## SELF-DIRECTED EDUCATION & CONTINUOUS LEARNING

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**1,000+ hours** of self-directed engineering study, built on primary documentation, open-source codebases, and structured curricula rather than a formal CS degree.

- **FreeCodeCamp** — Python Foundations (completed structured curriculum)
- **Harvard cs50** - Introduction to computer science
- **Vandad Nahavandipoor** — Advanced Dart & Flutter
- **Ben Dicken** — Computer Science & Software Engineering Fundamentals

- **Ongoing Research** — AI Automation and Cognitive Agent Frameworks, following Kevin Stratvert's practical build-alongs alongside direct study of CrewAI, LangGraph, and open-source agent-framework documentation